MGS613: Database Management Systems

Final Project

August 2025

Deva Sonar
Dora Lei
Pranali Koshti
Zachary Snow

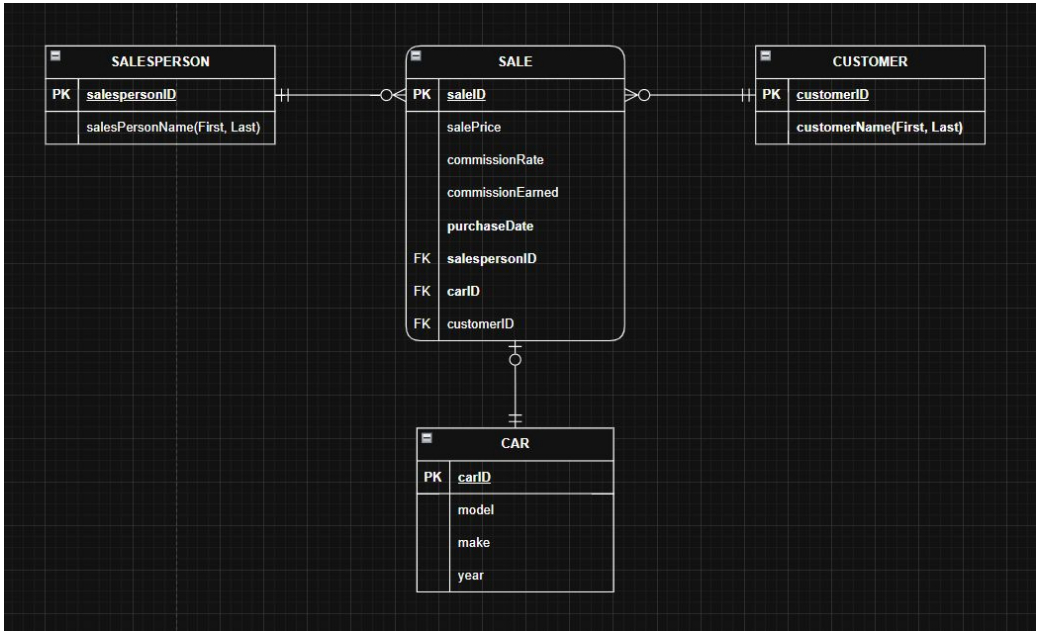
Purpose of the Project

- To build a relational database to track sales, customers, and salesperson performance.
- Identify best-selling vehicles and sales trends
- Track customer behavior (repeat purchases, preferences)
- Analyze salesperson performance (sales & commissions)

Business Rules:

- Each car can only be sold once.
- Each car sale is handled by only one salesperson.
- A customer doesn't have to buy a car but they can buy more than one car over time.
- A salesperson can be registered before they make any car sales.
- A salesperson can handle many car sales.
- Each sale record includes the sale price.
- Commission rate and commission earned for the salesperson is optional.
- All cars are new and not used.

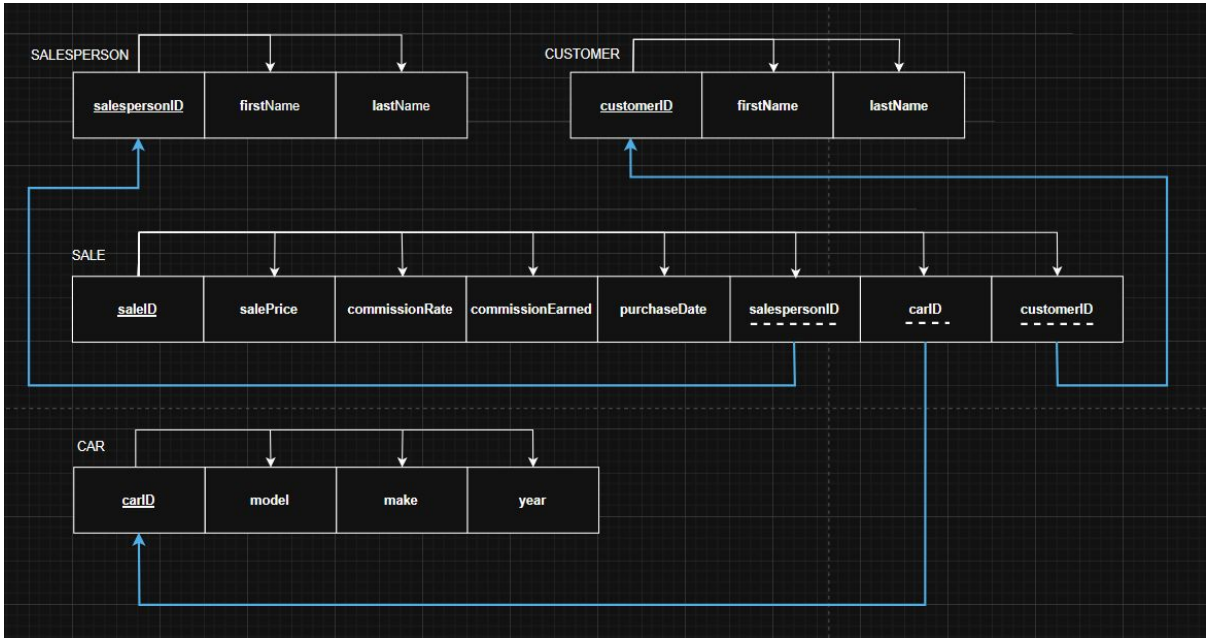
Entity - Relationship Diagram:



Data Dictionary:

Column Name	Data Type	Description
carID	int	Unique ID for each vehicle
make	text	Vehicle manufacturer
model	text	Vehicle model
year	int	Year of manufacturer
salePrice	float	Cost of car in USD
saleID	int	Unique ID for each transaction
salespersonID	int	Unique ID for each salesperson
salesPersonName	text	Salesperson name
commissionRate	int(optional)	Commission rate (%) for the sale
commissionEarned	int(optional)	Commission earned
purchaseDate	date	Date of transaction
customerName	text	Unique ID for each customer

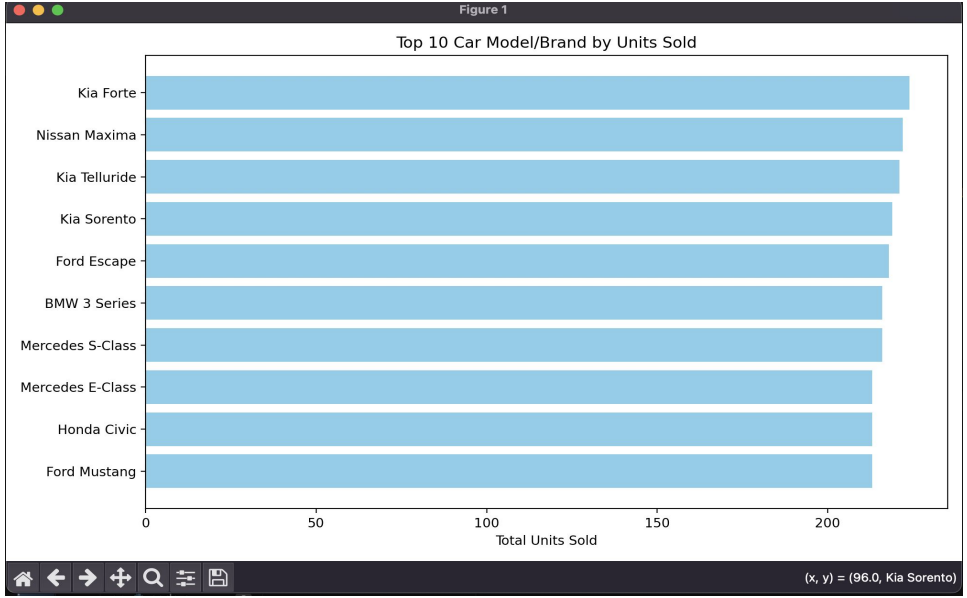
Relational Schema Diagram:



- The primary tables would be CAR, CUSTOMER, and SALESPERSON.
- The primary key for these tables are carID, customerID, and salespersonID.
- An associative entity called SALE with its primary key as saleID to link the car, customer, and company.
- There will also be three foreign keys in the SALE table.
- There will be a carID foreign key to link the sale to a specific car sold, a customerID foreign key to link the sale to the customer making the purchase, and a salespersonID foreign key to link the sale to the salesperson who helped.

Analysis

1. **Sales Trend:** Which car model/brand is bought the most?

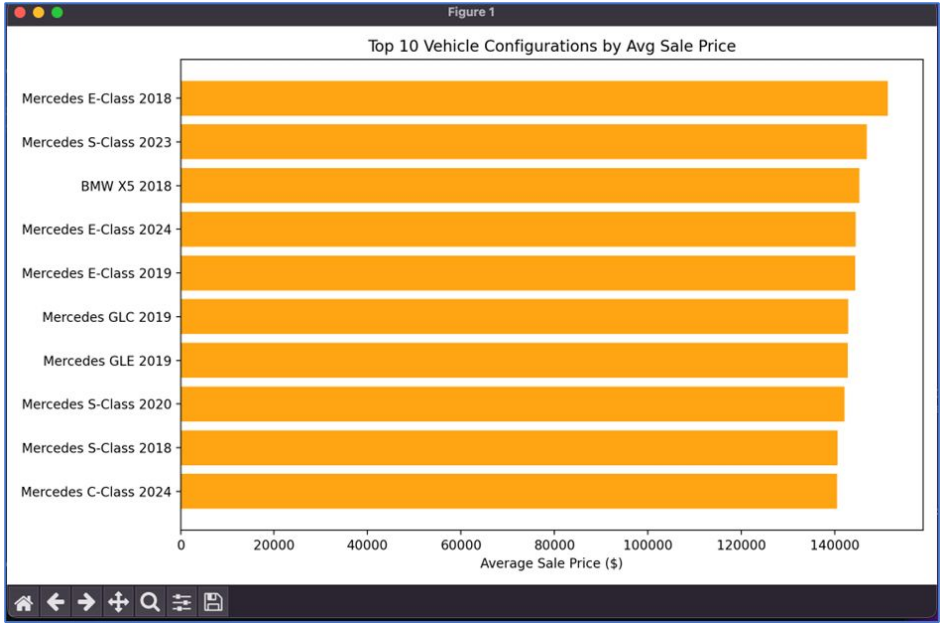


CODE:

```
SELECT
   c.CAR_MAKE,
   c.CAR_MODEL,
   COUNT(s.SALEID) AS Total_Sold
FROM SALE s JOIN CAR c
ON s.CARID = c.CARID
GROUP BY  c.CAR_MAKE, c.CAR_MODEL
ORDER BY Total_Sold DESC;
```

Analysis

2. **Sales Trend:** What is the average sale price of a vehicle based on their model, make and year?

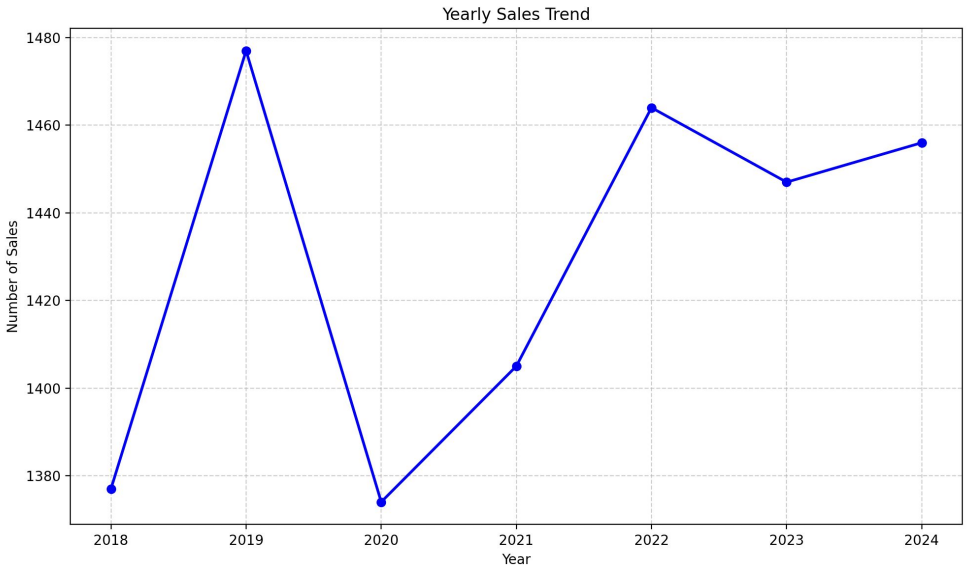


CODE:

```
SELECT
   c.CAR_MAKE,
   c.CAR_MODEL,
   c.CAR_YEAR,
ROUND(AVG(s.SALEPRICE), 2) AS
Avg_Sale_Price
FROM SALE s
JOIN CAR c ON s.CARID = c.CARID
GROUP BY c.CAR_MAKE, c.CAR_MODEL,
c.CAR_YEAR
ORDER BY Avg_Sale_Price DESC;
```

Analysis

3. **Sales Trend:** Sales trend over time (Year X count(sales))

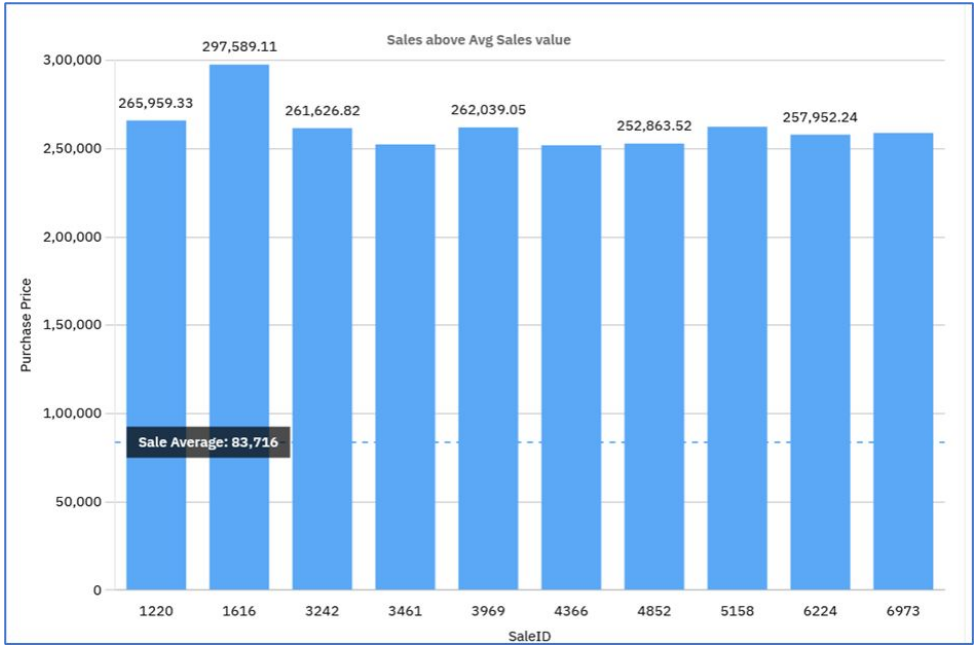


CODE:

```
SELECT
   TO_CHAR(PURCHASEDATE, 'YYYY')
AS year,
   COUNT(*) AS sale_count
FROM SALE
GROUP BY TO_CHAR(PURCHASEDATE,
'YYYY')
ORDER BY year;
```

Analysis

4. **Sales Trend :** What sales were above the average sale price?

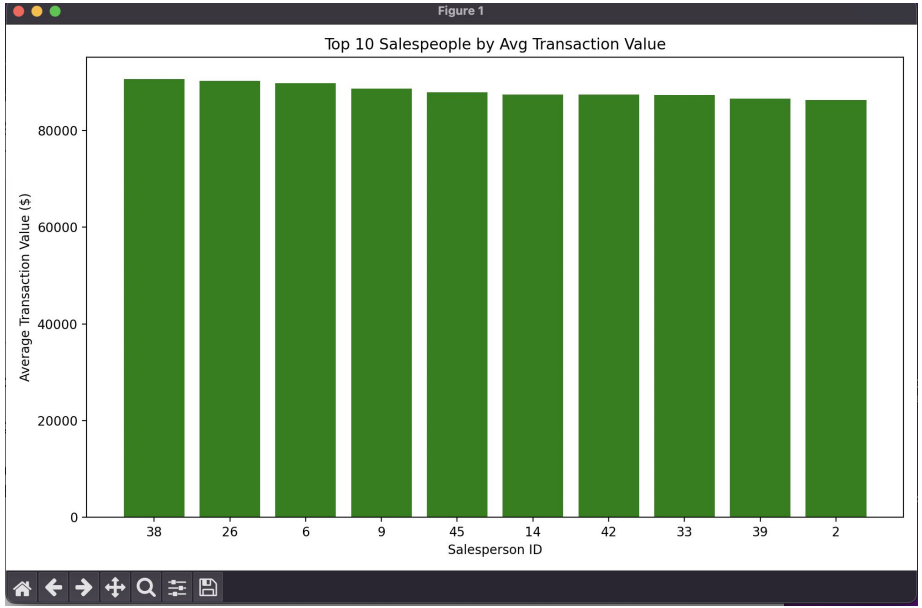


CODE:

```
SELECT SaleID,
TO_CHAR(AVG(Saleprice),'$999,999,999.
99') AS Avg_SalePrice
FROM Sale
GROUP BY Saleid
HAVING AVG(Saleprice) >
(SELECT AVG(Saleprice)
FROM Sale)
ORDER BY AVG(Saleprice) desc
FETCH first 10 rows only;
```

Analysis

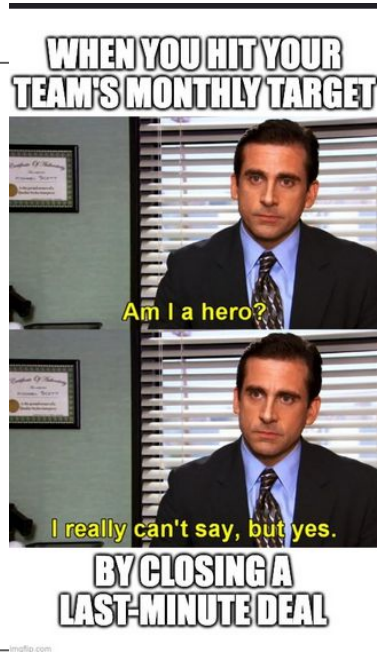
1. **Salesperson Performance:** What is the average sale price per transaction for each salesperson?



CODE:

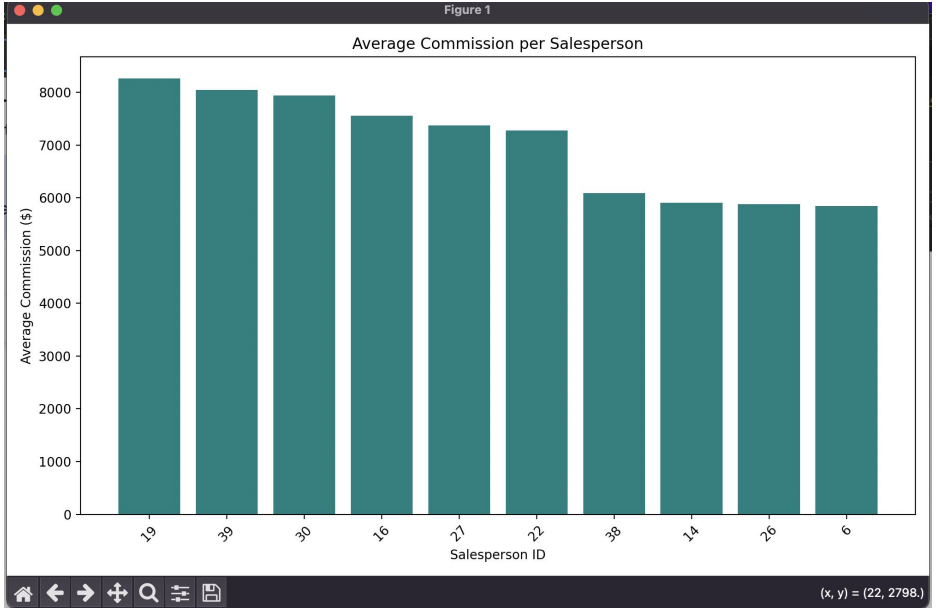
```
SELECT
SALESPERSONID,

TO_CHAR(AVG(SALEP
RICE), '$999,999,999.99')
AS
Avg_Transaction_Value
FROM   SALE
GROUP BY
SALESPERSONID
ORDER BY
Avg_Transaction_Value
DESC;
```



Analysis

2. **Salesperson Performance:** What's the average commission per salesperson?

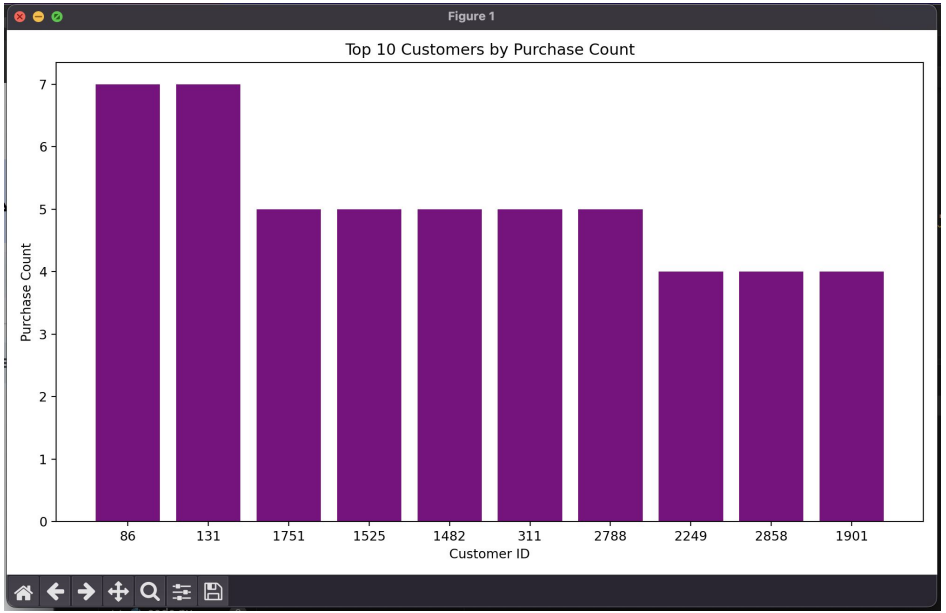


CODE:

```
SELECT
  SALESPERSONID,
  ROUND(AVG(COMMISSIONEARNED), 2)
AS AVG_COMMISSION
FROM SALE
GROUP BY SALESPERSONID
ORDER BY AVG_COMMISSION DESC;
```

Analysis

1. **Customer Behavior:** Which customers have made multiple purchases over time?



CODE:

```
SELECT
   CUSTOMERID,
   COUNT(*) AS Purchase_Count
FROM SALE
GROUP BY CUSTOMERID
HAVING COUNT(*) > 1
ORDER BY Purchase_Count DESC;
```

References:

1. Suraj520. "Car Sales Data." Kaggle
https://www.kaggle.com/datasets/suraj520/car-sales-data.

2. Anjali Prajapati307. "USA Car Sales Dataset 2018-2024." Kaggle
https://www.kaggle.com/datasets/anjaliprajapati307/usa-car-sales-dataset-2018-2024 KaggleKaggle

Thank You!